

COMPREHENSIVE CORPORATE BLUEPRINT

Ahmedabad Sustainable Scrap & Recycling Facility (ASRF)

An exhaustive, data-driven industrial blueprint establishing a multi-phase, low-risk metal and polymer recycling operation near Ahmedabad GIDC hubs.

PROJECT PROMOTER

Lead Entrepreneurial Group

PRIMARY FACILITY ZONE

Ahmedabad Peripheral Industrial Ring

PHASE 1 CAPITAL PLAN

₹5.2 Lakhs (Reserve-Locked Startup)

TARGET DEBT/SUBSIDY LEVEL

PMEGP Term Loan (₹20 Lakhs)

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■ 1. Executive Summary

The recycling and industrial scrap management sector in India has crossed a critical inflection point. Historically dominated by fragmented, unorganized players operating out of unlicensed yards, the industry is rapidly transitioning toward formalization. This shift is driven directly by new environmental laws, strict GST controls, and secondary steel manufacturing requirements that demand certified, high-grade scrap inputs.

The proposed **Ahmedabad Sustainable Scrap & Recycling Facility (ASRF)** represents a strategic opportunity to capture market share within one of India's most highly dense industrial casting and foundry states. The facility is modeled on a rigorous **"Capital Reservation Philosophy"**. Instead of investing heavy capital upfront on massive machinery and industrial land, ASRF initiates operations in Phase 1 as an **Asset-Light Aggregator & Sorting Center** with a small budget (₹5.2 Lakhs).

This phased launch validates localized supply lines, secures municipal and environmental clearances, and builds a stable customer portfolio using low overheads. Once cash flow matches operational targets, ASRF moves into Phase 2 by utilizing government-subsidized debt (PMEGP & CGTMSE) to purchase automated vertical/horizontal baling and shredding equipment, scaling the monthly throughput from 6 Tonnes to 40+ Tonnes.

Core Business Intent

By supplying clean, graded, and highly densified ferrous metals (steel, iron) and high-value non-ferrous metals (copper, aluminum, brass) directly to secondary steel producers and casting foundries in Vatva, Naroda, and Odhav GIDCs, ASRF establishes a reliable, recurring B2B trading framework.

■ 2. Future Potential of the Industry

India's national recycling industry is riding structural tailwinds that guarantee long-term volume growth. The domestic metal recycling market, valued at ****USD 11.4 Billion in 2024****, is projected to grow at a Compound Annual Growth Rate (CAGR) of ****6.8% to 7.3%****, reaching ****USD 21 Billion by 2030****.

| Key Structural Growth Drivers

- **National Steel Policy (NSP):** The government’s mandate aims to increase domestic steel manufacturing capacity to 300 million tonnes by 2030. To meet these targets without causing excessive carbon footprints, secondary steel manufacturers must utilize recycled steel scrap in Electric Arc Furnaces (EAF) and Induction Furnaces (IF).
- **The "Green Steel" Carbon Directive:** Global environmental standards, notably the European Union's Carbon Border Adjustment Mechanism (CBAM), penalize high-carbon steel exports. Indian steel exporters are rapidly converting their smelting systems to run on scrap steel inputs, which reduce raw emissions by 58% compared to virgin ore processing.
- **The Vehicle Scrappage Policy:** The phased ban on old, unfit private and commercial vehicles (older than 15–20 years) creates a continuous domestic supply of low-cost, high-grade automotive steel and copper scrap, driving prices down for recyclers.
- **The EPR Credit Market Explosion:** The Central Pollution Control Board’s digital registry forces major manufacturing brands to purchase Extended Producer Responsibility (EPR) credits from authorized recyclers, generating a digital, high-margin secondary income.

| 2.1 Urban Demolition, Infrastructure Upgrades & Sourcing Opportunities

To secure high-margin sourcing streams outside of competitive industrial workshop yards, ASRF will establish dedicated sourcing channels mapped to major municipal redevelopments, infrastructure expansions, and building demolition zones in the greater Ahmedabad region:

Identified Growth & Demolition Zone	Active Projects (2026 Timeline)	Scrap Type Generated (High-Value Targets)	ASRF Sourcing & Monetization Strategy
GIFT City Expansion &	• GIFT City physical expansion doubling its layout area by 2030.	• High-density construction structural steel girders and	• Partner with GMRC (Gujarat Metro Rail Corporation) tier-1 construction

Identified Growth & Demolition Zone	Active Projects (2026 Timeline)	Scrap Type Generated (High-Value Targets)	ASRF Sourcing & Monetization Strategy
Riverfront	- Sabarmati Riverfront 4.5-km Gandhinagar extension. - Approved Phase 2B elevated metro spur line linking GIFT City to Shahpur.	- sheet off-cuts. - Heavy armored electrical copper cables. - Temporary metal support shuttering and structural scaffolding frames.	- subcontractors for daily site scrap clearances. - Establish localized collection yards near Gandhidham outskirts to feed Gandhinagar projects directly.
AUDA Town Planning (TP) Corridors	Town planning road widening and anti-encroachment clearance drives in **Kalol, Thaltej, Sola, and Gota**.	- Corrugated iron roofing panels, metal shop front shutters, and structural framing sections. - Mild steel rebar from demolished concrete shops.	- Establish long-term supply arrangements with AUDA-enlisted private demolition contractors. - Buy mixed C&D (Construction & Demolition) metal scrap in bulk at steep discounts (up to 30% below retail Loha rate).
Old Walled City Demolitions	Ahmedabad Municipal Corporation (AMC) unsafe structural clearances and building redevelopments in **Shahpur, Kalupur, and Dariapur**.	- Thick Victorian-era cast-iron municipal water pipes. - Heavy load-bearing structural cast-iron pillars and columns. - Industrial-grade brass plumbing fixtures and old high-gauge copper wiring.	- Deploy specialized mobile gas-cutting teams to clear old properties. - Sort high-purity historical copper and cast iron on-site, commanding immediate premiums from Vatva foundries due to its clean chemical structure.
GIDC Old Yard Conversions	Defunct chemical, textile, and engineering mill conversions to modern warehousing blocks in **Vatva and Naroda GIDC**.	- Heavy obsolete machinery frames and solid cast-iron housings. - Steel portal structural frames from demolished factory sheds. - Heavy copper windings from decommissioned transformer substations.	- Offer "Complete Yard Clean-out Services" to factory owners. - Use Oxygen-Acetylene torch cutting to reduce structural portal beams down to standard **2-foot furnace-ready sections**, which command a **12% price premium** from secondary mills.

2.2 Dholera Semiconductor & Olympic Enclave Infrastructure Sourcing

The expansion of heavy industrial and civic construction in Gujarat opens up high-tonnage scrap channels. ASRF will establish an Infrastructure Scrap division to service two major multi-billion developments:

- Dholera SIR Semiconductor Fab (Tata-ASML Partnership):** The \$11 Billion commercial fabrication unit requires massive steel portal structures, heavy chemical delivery pipes, and cleanroom installation scaffolding. ASML's deep ultraviolet lithography tool deliveries beginning late 2026 drive fast-track construction, generating substantial steel and non-ferrous cabling waste.

- **Ahmedabad 2030 Commonwealth Games & 2036 Olympic Enclave (Motera):** Large-scale stadium construction at the Sardar Patel Sports Enclave creates thousands of tonnes of structural scrap steel rebar, thick copper power cables, and metal staging plates.

Infrastructure Sourcing Yield Advantage

Scrap from industrial fabs and sports arenas features high-grade, chemical-certified structural steel. Sorting and processing this scrap into 2-foot furnace-ready dimensions commands a **₹4,000 to ₹6,000 per Tonne** premium from secondary smelting induction furnaces.

3. Location & Zoning Strategy

Selecting the correct operational site is a critical decision for starting small-to-medium without burning excessive capital. The site must satisfy three distinct trade-offs: low monthly rental cash outflow, unrestricted heavy-vehicle access, and an accommodating GPCB (Gujarat Pollution Control Board) regulatory environment.

Below is an exhaustive comparative analysis of the primary GIDC clusters in and around the greater Ahmedabad industrial belt, rated by suitability for a startup aggregator/processor model:

Industrial Zone & Est. Rent (1,500 sq ft)	ASRF Suitability Rating	Key Sourcing & Operational Pros	Compliance & Operational Cons	Strategic Recommendation
Vatva GIDC Outskirts Rent: ₹25,000 - ₹35,000 / mo	7.5 / 10 ★ Moderate	- Immediate access to 50+ local castings foundries and steel buyers. - Virtually zero logistics cost for delivering sorted scrap batches.	- High plot congestion limits growth. - Strict GPCB air/water pollution monitoring under existing estate classifications.	Highly suitable for a pure sorting/trading branch, but high overhead makes a startup layout risky.
Odhav / Kathwada GIDC Rent: ₹22,000 - ₹30,000 / mo	8.0 / 10 ★★ Highly Viable	- Heavy concentrations of sheet-metal fabrication and engineering workshops. - Excellent day-1 localized scrap supply lines.	- Older GIDC infrastructure leads to heavy traffic congestion. - Strict municipal daytime vehicle restriction hours.	Strong contender if securing an outskirts plot on Kathwada fringes where space limits are slightly more flexible.
Sarkhej-Changodar Expressway Junction Rent: ₹20,000 - ₹28,000 / mo	9.8 / 10 🏆 Overall Winner	- Sits directly at the start of the NH-751 Ahmedabad-Dholera Expressway. - Immediate access to SP Ring Road for quick heavy truck transport to the northern Motera Sports Enclave. - Spacious land layouts outside civic municipal zone limits.	Higher demand for land plots near the expressway start is increasing leasing premiums.	Optimal location for the new Infrastructure Sourcing Division. Allows fast, unrestricted transport from both Dholera and Motera.

Industrial Zone & Est. Rent (1,500 sq ft)	ASRF Suitability Rating	Key Sourcing & Operational Pros	Compliance & Operational Cons	Strategic Recommendation
Chhatral GIDC (Gandhinagar Belt) Rent: ₹15,000 - ₹22,000 / mo	9.0 / 10 ★★★★★ High Potential	- Massive local casting foundry and steel rolling mill cluster (immediate buyers). - Substantially cheaper land rents (₹10-₹14/sq ft) and low structural compliance barriers.	- High density of local competitive scrap brokers and unorganized buyers. - Farther from Dholera SIR logistics routes.	Excellent balance of local foundry demand. Ideal location to start small, but less optimal for Dholera infrastructure logistics.
Kadi GIDC (Mehsana Outskirts) Rent: ₹14,000 - ₹20,000 / mo	8.0 / 10 ★★★ Highly Viable	- Cheapest rental threshold in the greater region. - Favorable GPCB regulatory landscape. - Massive industrial base (cotton, engineering, casting units).	Distance (45-50 km from central Ahmedabad) makes small-volume trade economically unviable.	Excellent for bulk aggregator models. Only select if shipping in consolidated full 15-Tonne truckloads.
Sanand GIDC Outskirts Rent: ₹30,000 - ₹40,000 / mo	5.0 / 10 ✖ Avoid for Startup	Clean, premium automotive scrap (stamping waste, high-purity off-cuts).	- Prohibitively expensive land rents and high lease deposit requirements. - Scrap contracts require participating in complex, high-capital corporate bidding.	Unsuitable for new entrepreneurs looking to conserve capital. Target as a sourcing lead later.

ASRF Siting Recommendation Matrix

For a capital-sensitive entrepreneur who wants to start small and mitigate down-side risks, the analysis yields two optimal pathways:

- 1. **Option A (Infrastructure Logistics Corridor - Sarkhej-Changodar Junction):** Secure a low-cost yard near the NH-751 Ahmedabad-Dholera Expressway start. This provides direct access to Dholera chip-fab steel clearing projects and SP Ring Road for Motera stadium deliveries. Keeps transit times low and bypasses city limits.
- 2. **Option B (Local Foundry Integration - Chhatral GIDC):** Rent a compact 1,500 sq ft industrial shed in Chhatral GIDC. Sourcing is active, overhead is low, and you can sell your sorted steel/iron scrap directly to the rolling mills next door, eliminating heavy transport outlays.

■ 4. National & Local Competitors

Entering this space requires understanding the operational structures of both national corporate leaders and organized local recycling groups.

| National Corporate Pioneers

- **Tata Steel Recycling (Tata FerroBaled / FerroShred):**

Model: Operates large, highly automated, multi-acre shredding hubs (such as their flagship facility in Rohtak, Haryana). Sourced scrap is crushed, processed, sorted magnetically, and compressed into uniform, high-density blocks branded as **FerroBaled**. Sourcing is managed digitally via their **FerroHaat** B2B app.

Takeaway for ASRF: We cannot compete on high-tonnage shredding. However, we can act as a feeder system, using manual sorting to supply high-purity scrap to secondary smelting hubs that corporate players cannot service flexibly.

- **Mahindra Accelo (CERO Recycling):**

Model: A joint venture with government-owned MSTC. Cero is India's leading Registered Vehicle Scrapping Facility (RVSF). They focus on extracting metals, rubber, glass, and copper wiring from end-of-life cars and commercial vehicles.

Takeaway for ASRF: Cero focuses strictly on automotive scrapping. We target localized industrial fabrication scrap, steel shavings, and packaging materials that offer higher turnover speeds and lighter regulatory pipelines.

- **CMR Green Technologies & Gravita India:**

Model: CMR is the undisputed national leader in aluminum casting alloy recycling, while Gravita specializes in lead and aluminum battery smelting.

Takeaway for ASRF: Illustrates the massive profit potential of specialized non-ferrous metals. ASRF will prioritize separating and grading copper and aluminum, as these represent only 15% of our raw volume but will contribute to 45% of our net margins.

| Local Gujarat Competitors

- **Kutch Green Recycling (Gandhidham / Anjar):** Large, port-adjacent sorting and processing yards. Focuses primarily on heavy maritime and structural steel scrap from ship-breaking yards.

| Top 3 Regional Competitors & Revenue Profiles

To secure market share, ASRF must strategically navigate around the top three dominant local/regional competitors operating in the greater Ahmedabad-Gandhinagar industrial belt. Each of these players has structural, regulatory, or operational vulnerabilities that ASRF is specifically engineered to exploit:

Competitor & Operational Base	Approximate Annual Revenue	Core Operational Model	Vulnerability (Where They Struggle)	ASRF Entry Playbook (How to Beat Them)
MKP Kataria Metals Odhav GIDC, Ahmedabad	₹120 Cr – ₹150 Cr (USD 14-18M)	Highly organized regional aggregator and licensed Registered Vehicle Scrapping Facility (RVSF). Clears heavy auto dismantling and plant machinery.	• Rigid, high-CapEx infrastructure with huge monthly overheads. • Completely ignores small-to-medium fabrication shops and localized construction waste because their logistics require minimum 5-Tonne pickups to be profitable.	• ASRF targets micro-sourcing leads (under 500 kg per pickup) using highly flexible, per-trip Tata Ace networks. • Offer immediate daily cash payouts, which their corporate treasury cannot clear through slow multi-tier bank approvals.
Siddhi Vinayak Iron & Steel Vatva GIDC, Ahmedabad	₹60 Cr – ₹80 Cr (USD 7-10M)	Aggressive trading firm focusing heavily on mild steel rebar and heavy melting scrap (HMS 1 & 2) for direct foundry feeding.	• Operates almost entirely in undocumented cash-based sourcing networks. • Fails to supply corporate buyers with verified GST Input Tax Credit (ITC) compliance documentation, exposing foundries to tax auditing penalties.	• ASRF operates with 100% tax compliance (GST Reverse Charge Section 9(4)). • Secure corporate foundries willing to pay a 5% to 8% margin premium strictly to get audit-proof, legal GST tax invoicing and claim their full ITC benefits safely.
Alang Metal Scrap Traders Changodar GIDC Branch	₹90 Cr – ₹110 Cr (USD 11-13M)	Branch of ship-breaking scrap brokers from Alang Port. Sells heavy structural steel plates, thick pipes, and copper armature wire from decommissioned vessels.	• Heavily exposed to international shipping scrap price fluctuations. • Focuses only on massive dispatches (minimum 15-Tonne dispatches), causing extremely slow local delivery times for nearby foundries.	• Act as a local buffer warehouse, supplying nearby Chhatral/Vatva foundries with flexible, same-day 2-Tonne scrap deliveries. • Shield local smelters from international pricing shocks by holding consistent physical buffer inventory on-site.

4.1 Regional Competitive Vulnerabilities & Strategic Playbook

To successfully enter the market, ASRF does not need to go head-to-head on pricing. Instead, we capitalize on the operational, financial, and compliance weaknesses of the existing traditional operators in each region:

Target Region	Existing Competitor Profile	Their Vulnerability (Where They Struggle)	ASRF Strategic Playbook (How to Win)
Vatva GIDC Outskirts	Densely packed unorganized traditional scrap traders (*kabadiwalas*).	• Tax Non-Compliance: Almost 90% operate strictly in undocumented cash, failing GST rules. • GST ITC Reclaim Losses: Smelting foundries face severe tax auditing penalties because these traders cannot issue legal input-tax-credit (ITC) papers.	• 100% Tax Integrity: Provide foundries with automated, digital GST-compliant weight slips and legal reverse-charge invoices (Section 9(4)). • Foundries will pay a 5% to 8% margin premium just to secure legal, audit-proof scrap.
Odhav / Kathwada GIDC	Mid-scale generic scrap sorting yards.	• Eyeball Grading Errors: They sort alloys using manual guesswork, often pricing high-purity copper/stainless steel off-cuts at flat low-grade rates. • Fabrication shops feel cheated by flat-rate scrap buyouts.	• Precision Grading: Deploy affordable chemical/spark metal testing kits on-site. • Offer fabrication shops precise, chemistry-based pricing for their high-grade metal stampings, instantly capturing their loyalty away from traditional buyers.
Changodar Hwy Corridor	Large-scale heavy structural steel brokers.	• Logistical Rigidity: They only deploy large 10-Tonne multi-axle trucks. • They completely refuse to service small/medium fabrication shops with daily scrap outputs under 500 kg because the logistics cost makes it unviable.	• Micro-Logistics Route: Form a dedicated "Tata Ace" pickup fleet. • Offer flexible, daily doorstep collections for small fabricators. Consolidate their small outputs at our yard into profitable 15-Tonne bulk dispatches.
Chhatral GIDC	Aggressive mill-aligned commission agents.	• Extreme Cash-Flow Illiquidity: Their working capital is permanently trapped in the long 45-60 day credit terms demanded by local rolling mills. • As a result, they frequently delay cash payments to their raw scrap suppliers.	• Liquid Invoice Factoring: Register ASRF under MSME Udyam and leverage digital invoice discounting platforms (TReDS). • Immediately convert foundry receivables into cash within 48 hours, allowing us to pay our scrap suppliers on the spot. Sourcing will naturally flow to us.
Kadi GIDC	Large mixed agricultural, cotton, and engineering yards.	• Product Contamination: They mix clean industrial fabrication scrap with general rural, plastic, and municipal wastes. • Mills often reject their entire truckloads due to high chemical impurity levels.	• Strict Industrial-Only Focus: Enforce zero-contamination policies at the yard. • Bale sheet stamping off-cuts separately from generic scrap metal. Secure "Preferred Vendor Status" with premium casting foundries.

■ 5. Legal Compliance Matrix

To avoid punitive GPCB shutdowns and ensure clean invoicing with corporate buyers, ASRF must secure the following licenses and certifications in a structured timeline:

Municipal Trade License			
License Name	Issuing Authority	Operational Necessity	Associated Fee / Cost
Udyam MSME Registration	Ministry of MSME (Govt. of India)	Unlocks access to collateral-free PMEGP/CGTMSE loans, 7% interest subsidies, and protection against delayed corporate buyer payments.	Free (Online Portal)
GSTIN Registration	Goods & Services Tax Department	Mandatory for B2B trade. Metal scrap falls under specific **Reverse Charge Mechanism (RCM)** rules (Section 9(4)), requiring careful input tax credit management.	Free (CA charges approx. ₹3,000)
GPCB Consent to Establish (CTE) & CTO	Gujarat Pollution Control Board	Required for municipal yard operations. Metal aggregation/manual sorting is categorized under **Green/Orange** , demanding simple run-off barriers.	₹15,000 - ₹35,000 (Based on Capex)
Legal Metrology Weighing Certificate	Department of Legal Metrology, Gujarat	Mandatory, legal calibration of industrial digital scales and weighbridges to prevent fraud. Subject to yearly renewal inspection.	₹5,000 - ₹8,000 per scale
Ahmedabad Municipal Corp. (AMC) / AUDA	Permits commercial waste accumulation and vehicle storage within local boundaries.	₹4,000 / annually	
DISH Factory License	Directorate of Industrial Safety & Health	Required only if operations scale to >10 workers utilizing electrical machinery (Phase 2 balers/shredders).	₹10,000 (Phase 2 only)

■ 6. Day-1 Sourcing Channels

Securing a continuous, daily input of raw scrap is the most critical survival factor for a startup recycling yard. ASRF will establish relationships with three primary sourcing channels on Day 1:

A

Automotive Workshops

Sourcing: SG Highway & Sanand Outskirts

Establish direct pickup contracts with the cluster of multi-brand car workshops, crash repair centers, and Suzuki/Tata automotive component stampers located near Sanand and SG Highway. Sourced materials consist of valuable thin mild steel panels, premium chassis structural steel off-cuts, and high-value copper wire harnesses.

B

Engineering & Fabrication

Sourcing: Kathwada, Odhav & Naroda GIDC

Partner with small to mid-scale sheet-metal punching units, turning workshops, and CNC machining centers. Sourced materials consist of highly consistent mild steel scrap sheet cut-outs, high-grade boring shavings, stamping leftovers, and aluminum sheet off-cuts. Immediate cash settlements ensure rapid loyalty.

C

Infrastructure & Demolition

Sourcing: SP Ring Road & GIFT Corridor

Form alliances with civil demolition contractors and metro transit sub-contractors on the Ahmedabad-Gandhinagar corridor. Sourced material consists of heavy, structural mild steel I-beams, TMT reinforcement rebar, thick cast-iron municipal pipes, and structural scaffolding tubes. High volume with maximum trade spreads.

D

Telecom & Electrical Networks

Sourcing: SP Ring Road Tower Grid

Partner with cell tower infrastructure operators and local electrical repair sub-contractors. Sourced materials consist of premium high-purity copper earthing plates, heavy braided copper grounding wire, aluminum heat sinks, structural angle-iron frames, and obsolete electrical switchgear.

E

Solar & Industrial Battery Pools

Sourcing: Charanka solar grid suppliers

Establish contracts with industrial battery replacement groups and regional solar farm maintenance teams near rural Ahmedabad boundaries. Sourced materials consist of expired heavy B2B lead-acid backup battery blocks, structural aluminum solar mounting frames, and copper battery link connectors.

■ 6.1 Non-Traditional & Unmonetized Sourcing Channels

A key operational leverage point for ASRF is sourcing high-purity, heavy scrap from institutional partners that traditional traders (*kabadiwalas*) fail to target or price correctly. Because these entities prioritize yard clearance and safety compliance over hard commodity pricing, ASRF can lock in high-grade metal assets at extreme price spreads:

Overlooked Institution / Location	Unmonetized Asset Profile (Sourcing Targets)	ASRF Strategic Sourcing Approach	Spread Advantage (Profit Optimization)
Hospitals & Healthcare Facilities	Decommissioned non-biohazardous assets: metal ward beds, wheelchairs, stainless-steel clinical tables, autoclave sanitizers, brass cylinder valves/regulators, and thick lead-shielding panels from X-ray room updates.	ASRF issues a formal "Safe Assets Clearance Certification", guaranteeing that we only collect non-infectious, certified non-biohazardous metal structures, completely removing the hospital's storage disposal liability.	Hospitals prioritize safety and prompt clearing. Metal is bought in bulk at a 35% discount compared to standard industrial fabrication scrap.
Schools, Colleges & Universities	Broken mild-steel framed desks, institutional kitchen aluminum vessels, large stainless-steel canteen sinks, obsolete physics/chemistry lab metal items (brass Bunsen burners, brass calipers, cast-iron retort stands).	Offer an "Annual Campus Salvage Clean-out Contract". ASRF's teams sweep clean the campus junk yards during summer/winter student holidays, presenting the college with clean municipal space and official compliance bills.	Trusts sell old assets at nominal book values to avoid storage clutter. High-volume clean aluminum and structural mild steel collected at flat minimum rates.
Traffic Control & Roads Department	Dismantled heavy galvanized-steel traffic posts, bent highway guardrail beams, municipal traffic light steel boxes, high-purity copper traffic controller junction boxes, and heavy cast-iron manhole covers.	Establish direct supply channels with the private subcontractors executing Smart-City roads upgrades on the SP Ring Road and peripheral highways, offering instant cash clearance on site.	Subcontractors view these as site waste. ASRF aggregates highly premium galvanized structural steel (which resists rust and commands a high melting value) without participating in long municipal auctions.
Bank Branch Refits & Strong-rooms	Thick structural steel vault doors, security grilles, old teller counters, armored safe panels, server room heavy racks, and massive server power copper cabling bundles.	Offer a "Zero-Cost Structural Extraction Service". Bank asset managers face high logistics costs to tear down heavy steel vaults. ASRF executes the extraction free of charge in exchange for keeping all structural metals.	Vault door steel represents extremely high-grade, thick industrial steel scrap. Monetized by selling directly to high-grade foundries at a 100% margin spread above extraction costs.
Telecom Towers &	Decommissioned network towers: copper grounding earthing plates,	Partner with cellular network maintenance subcontractors	Copper earthing plates are extremely thick, high-purity

Overlooked Institution / Location	Unmonetized Asset Profile (Sourcing Targets)	ASRF Strategic Sourcing Approach	Spread Advantage (Profit Optimization)
Data Hubs	heavy braided copper grounding wire, structural angle-iron frames, and expired lead-acid telecom battery banks.	covering the Ahmedabad-Gandhinagar grid, setting up automatic collection schedules for replaced batteries and ground wire strips.	electrolytic copper, which foundries buy at a 15% premium above standard thin copper wire scrap.

■ 7. Target Buyers & Off-Taker Networks

Rather than selling scrap back to other intermediaries (which dilutes profit margins by 5-10%), ASRF will sell processed and aggregated scrap directly to final B2B industrial consumers.

| Immediate High-Volume Buyers in Ahmedabad

- **Kamal Foundry & Shree Hans Alloys (Vatva & Chhatral GIDC):**

Material Required: High-purity sorted cast iron scrap, heavy melting steel, high-purity brass castings, and clean copper wire scrap.

Purchase Terms: Standard monthly supply agreement with 15-day credit invoicing once credentials are verified. Requires strict weight slips from government-calibrated scales.

- **Electrotherm India Limited & Shah Alloys (Palaj / Santej Outskirts):**

Material Required: Baled or shredded structural mild steel scrap, heavy melting scrap (HMS 1 & HMS 2) for induction furnace feed.

Purchase Terms: Bulk purchase orders with dynamic pricing indexed weekly to the regional Ahmedabad Loha scrap rate.

- **CMR Toyotsu Aluminum Recycling (Sanand GIDC - Toyota Joint Venture Hub):**

Material Required: Pure aluminum scrap off-cuts, aluminum boring shavings, solar mounting frame profiles, and auto sheet-metal scrap for secondary **ADC12 Aluminum casting alloy** production.

Purchase Terms: Scheduled weekly delivery dispatches. Instant digital GST invoicing with 10-day payment clearance via corporate banking lines.

- **Gravita India Limited (Gandhidham / Port Outskirts):**

Material Required: Obsolete industrial lead-acid backup batteries, clean lead sheets, and heavy lead scrap.

Purchase Terms: Consolidated dispatch loads (minimum 5-Tonne batches) under pollution control board transit manifesting rules. Immediate bank payment upon site intake chemical assay testing.

- **Welspun Power & Steel & GPT Steel (Anjar/Bhachau, Kutch):**

Material Required: Highly compressed, uniform steel and iron baled blocks.

Logistics Plan: Consolidated Phase 1/Phase 2 yard stocks are loaded onto flatbed dumper trailers (15-Tonne capacity) and dispatched twice monthly to maximize transport economics.

- **Adani Wilmar & local refining complexes (Mundra / Halol):**

Material Required: Obsolete structural plant components, retired heavy machinery cast-iron parts, heavy

electric motor copper windings, and stainless-steel processing pipes.

***Purchase Terms:** Strategic clearing contracts, cleared on-demand by ASRF on a per-tonne payment structure, bypassing local unorganized brokers.

■ 8. Day 1 Startup Checklist

To launch ASRF efficiently while keeping overhead minimal, we will execute the following day-1 operational checklist. This avoids wasted capital and ensures zero legal vulnerabilities from day one:

Action Item	Critical Requirement Details	Responsible Party	Capital Outlay Status
1. Lease Execution	Execute a commercial lease for a 1,500 sq ft secure walled plot or open shed in Changodar or Odhav outliers. Limit security deposit to 3 months.	Promoter Team	₹75,000 (3-Month Deposit)
2. Entity & GST Setup	Register the partnership deed, obtain the PAN card under the firm name, secure Udyam MSME, and apply for a current bank account and GSTIN.	Chartered Accountant	₹10,000 (Legal Fees)
3. Yard Security & Fencing	Install razor-wire boundary fencing, deploy a simple metal lock-up cabin for tools, and install 4 solar-powered CCTV cameras.	Local Contractor	₹50,000 (Materials & Labor)
4. Weighing Gear Procurement	Purchase one digital heavy platform scale (1,000 kg capacity) and one high-precision platform scale (100 kg capacity for copper/brass).	Industrial Scale Vendor	₹45,000 (Digital Calibration)
5. Weights Verification	Invite the local Legal Metrology Inspector to inspect, seal, and stamp the digital scales, securing the mandatory calibration certificate.	Legal Metrology Dept.	₹6,000 (Official Stamp Fee)
6. Sourcing Transport Setup	Secure a dedicated account with local Tata Ace / Chota Hathi cargo driver networks for flexible, per-trip scrap collection pickups.	Operations Manager	No upfront cost (Per-trip payout)

9. Extended Financial Projections

The financial model is designed to optimize cash conservation. Below is a granular, month-by-month financial projection for Phase 1 (Aggregator Phase) based on a target daily collection of ****240 kg**** of mixed metals.

Detailed Phase 1 Capex (₹5.2 Lakhs)

Investment Expense Item	Budget Required (INR)
Yard Security Deposit (3 Months)	₹75,000
Fencing & Secure Storage Cabin	₹50,000
Digital Platform Scales (Certified)	₹45,000
Legal Fees (GST, MSME, GPCB Consent)	₹50,000
Hand Tools, Cutters, & Safety Equipment	₹25,000
Initial Working Cash Buffer (3-day cycle)	₹2,00,000
Yard Signage, Fire Extinguishers, Basics	₹25,000
Total Phase 1 Startup Capex	₹4,70,000

Monthly Phase 1 Opex Breakdown

Monthly Operational Cost Item	Budget Outlay (INR)
Open Yard / Shed Rent (1,500 sq ft)	₹25,000
Labor Payroll (2 Segregators)	₹30,000
Tata Ace / Local Logistics (Per-trip basis)	₹22,000
Electricity, Water, & Basic Utilities	₹6,000

Monthly Operational Cost Item	Budget Outlay (INR)
Yard Maintenance & Equipment Fuel	₹7,000
Total Monthly Phase 1 Opex	₹90,000

3-Year Financial Scaling Target

- **Year 1 (Aggregator Phase):** Target Blended Daily Sourcing: 350 kg/day. Expected Annual Revenue: **₹42 Lakhs**. Expected Net annual profit (after Opex, COGS): **₹6.4 Lakhs**. (Initial CapEx recovered within 10 months).
- **Year 2 (Processor Transition - Funded by PMEGP):** Purchase horizontal baler (₹7.5L) and dual-shaft shredder (₹4.5L). Scale collection to 1,500 kg/day (37.5 Tonnes/month) by contracting fabrication factories in Vatva and Odhav. Expected Annual Revenue: **₹2.2 Crores**. Expected net annual profit: **₹32 Lakhs**.
- **Year 3 (Enterprise Grading Yard):** Relocate to a larger 5,000 sq ft Changodar facility. Target 3.5 Tonnes daily throughput. Expected Annual Revenue: **₹5.4 Crores**. Expected net annual profit: **₹88 Lakhs**.

■ 9.1 Risk-Shielded Asset Strategies & Joint Ventures

Heavy scrap processing machinery (hydraulic balers, alligator shears, crane loaders) represents a double-edged sword for a startup. They require significant upfront capital expenditure (₹12 Lakhs – ₹25 Lakhs) and suffer from rapid depreciation (typically 15% to 30% per annum under the Indian Income Tax Act). If raw sourcing volumes fluctuate, fixed machinery debt and depreciation will quickly drain working capital.

To eliminate this asset risk while retaining 100% of ASRF's growth potential, we will deploy four defensive capital protection strategies:

1

Toll Processing & Job-Work Partnerships

Outsourcing Capital-Heavy Compacting

Instead of purchasing a baler, ASRF will partner with under-utilized plastic recycling and metal fabrication units in Odhav and Kathwada. These yards have heavy-duty hydraulic baling presses running at under 40% capacity. We aggregate and sort raw metals at our open yard, then route them to their facilities under a *****Job-Work Toll Contract***** (paying a flat ₹800 – ₹1,200 per tonne processed).

Benefit: Completely bypasses the ₹12L baler CapEx, transforming fixed machinery debt into a variable per-tonne fee while ASRF retains 100% of sales branding and margins.

2

Distressed Asset Acquisitions

Buying Closed-Yard Liquidations

Many local scrap yards and fabrication workshops face closure due to cash flow insolvencies or tax non-compliance. ASRF will actively source second-hand machinery (platform scales, crocodile shears, wire strippers) from liquidation brokers and MSTC India public auction portals.

Benefit: Secure fully functional, certified industrial recycling machinery at ****40% to 60% below book value****, avoiding dealer markups and drastically accelerating capital payback.

3

Foundry Co-Location Joint Venture

Shared Machinery Integration

Partner with an existing, mid-scale melting foundry in Chhatral or Vatva GIDC that possesses under-utilized industrial weighbridges, overhead gantry cranes, and massive shearing machinery. ASRF co-locates our sorting operations inside their vacant boundary acreage.

Benefit: In exchange for a guaranteed first-right-of-refusal to purchase our sorted scrap at a fixed 3% discount (retaining their transport costs), they allow ASRF to use their cranes and heavy shears for free during their night shifts, reducing our Phase 2 CapEx to zero.

4

Operational B2B Leasing Models

Shifting CapEx to Tax-Deductible OpEx

For necessary material handlers and high-capacity balers, ASRF will reject outright purchasing. Instead, we will contract with B2B machinery lessors (such as Tata Capital B2B leasing or regional machinery distributors) under a 3-year operating lease.

Benefit: Preserves our cash reserve. Monthly lease payments are treated as 100% tax-deductible operational expenditure (OpEx), lowering net corporate tax liability without weighing down our balance sheet with depreciating assets.

■ 10. EPR Credit Monetization

Extended Producer Responsibility (EPR) represents a massive paradigm shift in Indian recycling monetization. The Ministry of Environment, Forest and Climate Change (MoEFCC) mandates that any brand using plastic packaging or manufacturing electronic goods (PIBOs - Producers, Importers, Brand Owners) must legally purchase EPR credits to offset their environmental footprint.

EPR Digital Credit Generation Pathway

1. **CPCB Registration:** ASRF registers as an authorized waste processor on the CPCB Portal, uploading our GPCB Consent to Operate (CTO) and weighing capacity metrics.
2. **Verification of Inflow:** As we collect and aggregate post-consumer hard plastics (PET, HDPE bottles) and electronic waste components, all purchases and sales are recorded via GST-compliant e-way bills.
3. **Digital Credit Issuance:** CPCB algorithmically issues 1 digital ****EPR Credit**** on the portal for every 1 Tonne of verified recycling processed.
4. **Credit Trading Board:** ASRF lists these credits on the CPCB trading board, where major corporate entities (such as Reliance, PepsiCo, ITC) bid and purchase them directly, yielding a pure profit margin of ****₹2,000 to ₹5,000 per Tonne**** without physical product delivery.

11. Risk Analysis & Mitigation

Every business faces unique market risks. Sourcing and trading metal and polymer scrap requires proactive hedging against local market disruptions:

Identified Business Risk	Impact Level	ASRF Defensive Mitigation Strategy
Scrap Commodity Price Volatility	High	Maintain a strict 3-Day Inventory Turnaround Cycle . Avoid holding scrap in the yard expecting speculative price increases. Immediate liquidation locks in the physical trading spread.
Local Sourcing Monopoly / Theft	Medium	Install solar-powered 24/7 CCTV surveillance cameras. Diversity supply lines by sourcing from 20+ micro-aggregators and workshop accounts, preventing dependence on single vendors.
Delayed Off-Taker Corporate Payments	Medium	Utilize MSME Udyam registration credentials. Under the MSME Act, all corporate buyers must legally settle invoices within **45 days** , failing which they face compound interest penalties.
Unexpected GPCB Compliance Actions	High	Design the yard layout with impermeable, concrete run-off basins. Avoid any acidic dismantling or chemical extraction processes in Phase 1. Limit operations strictly to dry, physical segregation.

11.1 Worst-Case Stress-Testing, Strategic Failure & Liquidation Playbook

To secure ASRF against systemic failures, we stress-test the business under extreme operational and market conditions, establishing pre-planned strategic mitigations:

Stress Scenario	Immediate Financial Impact	Pre-Planned Mitigation & Escape Route
Local Cartel Blockade / Sourcing Boycott Local scrap brokers form a cartel, refusing to supply ASRF or blockade transport trucks from Vatva.	Loss of Day-1 supply flow; yard overheads burn ₹90,000/mo without scrap revenue.	- Deploy our mobile pickup networks to northern regions (Gandhinagar/Mehsana TP schemes) to bypass local cartel areas. - Offer higher digital payment terms via TReDS invoice factoring, attracting micro-scrap collectors who need fast cash.
Global Steel/Copper Price Crash (40% drop in 1 week)	Instant inventory write-down of ₹2.5 Lakhs. Cash flow crunch.	Enforce an absolute 3-Day Holding Velocity Limit: scrap must leave the yard within 72 hours of purchase.

Stress Scenario	Immediate Financial Impact	Pre-Planned Mitigation & Escape Route
Inventory value crashes while holding 20 Tonnes of steel scrap in the yard.		Shift immediately to a back-to-back sorting model: secure off-taker purchase price locks before purchasing scrap from suppliers.
Dholera Construction Delay or Bidding Disqualification Tata-ASML project delayed by 12 months; ASRF Infrastructure Division fails to win bids.	Infrastructure Division overhead loss (₹1.5 Lakhs/mo); zero Dholera revenues.	- Pivot the division immediately to clear structural scrap from the Motera Sports Enclave projects. - Maintain a low fixed-cost cost structure: hire crane operators and scrap trucks on a per-diem basis rather than purchasing heavy hardware.
GPCB Zero Liquid Discharge (ZLD) Violation Notice Runoff water from outdoor scrap sorting yards triggers GPCB environmental audit warnings.	Threat of immediate factory closure or heavy daily compliance fines.	- Construct elevated canopy sheds over all sorting zones to keep rainwater isolated from stored metal. - Pre-install epoxy-coated concrete floors and chemical sand filters in the runoff basins to guarantee zero toxic seepage.
Liquidation Trigger (Go/No-Go Break Point) Cumulative net losses exceed ₹4 Lakhs or monthly volume drops below 2 Tonnes for 3 consecutive months.	Strategic decision to shut down operations and protect investor capital.	Capital Preservation Liquidation Protocol: - Liquidate all remaining scrap inventory to Vatva foundries within 48 hours at a 10% discount. - Sell the modular office containers, weighing scales, and sorting benches to recover estimated ₹2.5 Lakhs in asset capital. - Terminate the short-term land lease (requires 30-day notice) to prevent further rental burn. - Exit clean with over 50% of starting capital preserved.

■ 12. Global Sourcing & Scrap Import Mechanics

As ASRF scales into Phase 3, relying solely on local workshop scrap limits daily throughput. To reach industrial-scale viability, the facility will establish a structured international supply line to import high-purity metallic waste from the United States, the United Kingdom, and the European Union.

Gujarat offers a strategic geopolitical advantage for global scrap trade. The state features two of India's largest deep-water ports—**Mundra Port** (Adani Special Economic Zone) and **Kandla Port** (Deendayal Port Authority)—both of which are formally designated by the Directorate General of Foreign Trade (DGFT) as licensed clearing ports for metallic waste imports.

| DGFT Compliance & Exemption Pathways

Importing scrap into India is highly regulated to prevent the dumping of hazardous, explosive, or radioactive materials. ASRF will execute imports under the following statutory frameworks:

- **Pre-Shipment Inspection Certificate (PSIC) Exemption:** Under DGFT Public Notice guidelines, metallic scrap consignments imported from "Safe Regions" (specifically the US, UK, Canada, EU, Australia, and New Zealand) are **exempted from the requirement of a Pre-Shipment Inspection Agency (PSIA) certificate** at the port of origin.
- **Supplier Radiation Declarations:** To utilize the PSIC exemption, each imported container must be accompanied by a formal declaration from the scrap yard authority at the origin confirming that the consignment contains zero radioactive or explosive residues.
- **Mandatory Port Scans:** Upon arrival at Mundra or Kandla, containers undergo automated portal radiation monitor scans and physical container x-ray checks conducted by Indian Customs before final domestic clearance.
- **Customs House Agent (CHA) Integration:** Establish a dedicated account with a licensed CHA at Gandhidham to manage the filing of the Bill of Entry on the ICEGATE portal, ensuring immediate clearance to avoid container demurrage charges (which can run up to ₹15,000 per container per day).

■ 13. Secondary Smelting & Ingot Export Framework

Transitioning from a pure raw metal aggregator to a value-added manufacturer represents the ultimate margin-expansion strategy for ASRF. In Phase 3, we will install localized induction and rotary smelting furnaces to process raw sorted aluminum and copper scrap into standardized metallurgical ingots.

Value-Added Products

Instead of selling loose aluminum castings at low scrap value (₹140/kg), ASRF will smelt and alloy them into standard **ADC12 Aluminum Die-Casting Alloys** or **LM6 Aluminum Ingots** (valued at ₹210+/kg). These alloys are purchased in bulk by automotive OEM parts manufacturers in Sanand GIDC and international buyers in South-East Asia.

Export Incentives

By refining non-ferrous metals to 99.7% purity, ASRF gains access to high-margin export markets in Europe and China. Under the EU's Carbon Border Adjustment Mechanism (CBAM), European buyers pay a substantial pricing premium for **certified secondary (recycled) metal ingots** because they carry a fraction of the carbon footprint of virgin-smelted metals.

DGFT Advance Authorization & SION Guidelines

ASRF will utilize the ****Advance Authorization Scheme**** to import raw metallic scrap duty-free. Under this scheme, the government waives the basic customs duty on imported inputs, provided the material is processed and exported as finished metal ingots within 18 months. Operations must strictly adhere to the pre-fixed ****Standard Input-Output Norms (SION)****, which mandate precise conversion yield ratios (e.g., maximum permissible process melting loss of 4% for copper and 7% for aluminum scrap).

■ 14. Advanced Yard Sorting Technologies

As operational volumes cross 100 Tonnes per month, manual sorting by hand segregators becomes an efficiency bottleneck. Phase 3 of ASRF integrates automated sensor-based sorting systems to secure near-100% material purity:

1

X-Ray Transmission (XRT)

Deploys high-speed x-ray sensors to analyze the atomic density of incoming scrap on a conveyor belt. Instantly segregates heavy metals (copper, brass, zinc) from light aluminum alloys, sorting up to 5 Tonnes per hour with zero human labor.

2

Near-Infrared (NIR) Sorting

Essential for polymer sorting. Measures the light absorption spectrum of moving plastic fragments to instantly differentiate and sort HDPE, PET, PP, and PVC polymers, securing 99.8% pure polymer streams for GPCB/CPCB compliance.

3

LIBS Technology

Laser-Induced Breakdown Spectroscopy uses nanosecond laser pulses to vaporize micro-points of alloy scrap. Instantly reads precise chemical percentages (e.g., separating 6000-series from 5000-series aluminum), allowing premium alloy pricing.

■ 15. Environmental, Health & Safety (EHS) Protocols

Operating a heavy industrial recycling yard carrying toxic or flammable components presents serious compliance liabilities if unmanaged. ASRF enforces a rigorous, zero-compromise EHS framework:

| GPCB Water & Ground Pollution Barriers

To prevent toxic heavy metals and battery acids from leaching into the local Ahmedabad water table, the yard layout integrates the following physical structures:

- **Impermeable Concrete Containment Beds:** All battery storage, sorting, and cutting areas must feature a 6-inch thick, epoxy-coated concrete floor. This prevents any chemical spill or acidic battery fluid from making direct ground contact.
- **Spill Catchment Trenches:** The perimeter of the non-ferrous processing floor is bordered by chemical-resistant collection trenches leading to a central holding tank for neutralization processing.
- **Rainwater Run-off Isolation:** The yard features physical diversion bunds that separate clean storm rainwater from the scrap storage zone, preventing stormwater contamination in compliance with GPCB Consent rules.

| Worker Safety & Yard Operations

All operations under the DISH Factories Act regulations must enforce strict safety rules: steel-toed boots, heavy leather sorting gloves, eye-protection goggles for metal cutters, and designated, color-coded safety walkways that isolate workers from moving forklift trucks and baling areas.

■ 16. Local Union & Labor Dynamics

Navigating the operational realities of Ahmedabad's industrial GIDC belts requires managing localized labor structures and logistical unions:

| Managing Gujarat GIDC Transport & Labor Unions

Traditional estates like Vatva and Chhatral are heavily influenced by local transport associations and loader unions. If unmanaged, these unions can demand arbitrary loading rates or block outside logistics trucks from entering your yard. ASRF will implement three defensive strategies:

- **Direct Local Hiring:** Hire yard loaders and security staff from neighboring villages (e.g., near Changodar or Chhatral outskirts). This builds deep local community goodwill, neutralizing hostile union actions.
- **Registered MSME Supplier Protections:** Ensure all transport agreements are formalized with registered logistics providers under GST, utilizing statutory protection clauses to prevent illegal truck blockades.
- **Contract Labor Licensing:** Secure a formal Contract Labor License under the Contract Labour (Regulation & Abolition) Act, 1970, ensuring all temporary yard segregators are registered, covered under Employee State Insurance (ESIC), and paid standard minimum wages via digital bank transfer, eliminating underground labor disputes.

17. Strategic Summary & Go/No-Go Investment Verdict

Based on extensive regulatory, operational, logistical, and competitor modeling within the Ahmedabad-Gandhinagar industrial belts, this proposal offers a definitive structural analysis of the startup's viability.



Investment Verdict: GO (Highly Recommended)

Launching the ****Ahmedabad Sustainable Scrap & Recycling Facility (ASRF)**** is an exceptionally viable business venture. Sourcing and aggregating industrial metal and polymer waste is a high-spread, fast-liquidation trading activity. By utilizing the **Asset-Light Aggregator Model** in Phase 1, you limit your downside risk and keep your starting Capex under ₹5 Lakhs, while securing a highly structured operational foundation ready to scale into B2B manufacturing and international trade.

Average Monthly Operating Costs

Cost Overhead Category	Phase 1 Burn (INR/mo)	Phase 2 Scale (INR/mo)	Phase 3 Infra scale (INR/mo)
Yard / Shed Rental	₹25,000	₹65,000	₹1,20,000
Labor Payroll & ESIC	₹30,000	₹1,20,000	₹2,80,000
Logistics & Transport	₹22,000	₹95,000	₹2,50,000
Power & Heavy Utilities	₹6,000	₹45,000	₹95,000
Maintenance & Fuel	₹7,000	₹25,000	₹55,000
Total Monthly Burn	₹90,000	₹3,50,000	₹8,00,000

Liquidation & Cash Flow Playbook

To maximize operational efficiency and prevent cash lock-ups in heavy steel scrap piles, ASRF will execute three strict financial strategies:

- Strict 3-Day Holding Velocity Limit:** Enforce an absolute rule: scrap entering the yard must be graded, bundled/baled, and dispatched to buyers within 72 hours. This prevents physical inventory speculation and

ensures a continuous recycling cash loop.

- **Index-Synced Sourcing Pricing:** Never buy scrap at flat speculative rates. Sync your purchasing price lists daily with regional steel indices (e.g., Loha Mandi Gobindgarh and local Alang ship-breaking scrap price charts) minus a fixed 15% operational spread. This insulates ASRF from sudden global commodity price drops.
- **MSME TReDS B2B Cash Discounting:** Avoid traditional 45-day corporate credit payment terms. Register the company as an MSME on digital discounting networks (TReDS, M1Xchange, or RXIL). Instantly trade approved B2B invoices with partner banks to receive ****liquid cash in your account within 24 to 48 hours**** at a minimal bank discount fee of 1.5%.

17.1 Extended Go-Scale Target Model (₹10+ Crore Output)

By establishing the specialized Infrastructure Sourcing Division, ASRF transitions from local workshop aggregating to high-tonnage institutional contracts. This shifts the growth projections dramatically:

Growth Phase	Target Sourcing Volumes	Average Net Margin Spread	Annualized Revenue Target	Key Capital Sourcing Channels
Phase 1: Startup Aggregating (Months 1 - 6)	6 - 10 Tonnes/month (Workshop metal/PET)	12% - 15% (₹3,000/Tonne)	₹20 - ₹30 Lakhs	Self-financed (₹5.2 Lakhs reserves)
Phase 2: Automated Processing (Months 7 - 18)	30 - 50 Tonnes/month (Industrial baling)	15% - 18% (₹4,500/Tonne)	₹1.2 - ₹1.8 Crores	PMEGP + CGTMSE Mudra loans (₹20 Lakhs)
Phase 3: Dholera/Olympic Clearing (Months 19 - 36)	200 - 300 Tonnes/month (Structural scrap)	20% - 22% (₹6,000/Tonne)	₹10 - ₹12 Crores	Reinvested profits + private project financing lines